

Quiz 5: Quantum Information Processing Protocols

Question 1. Consider a quantum teleportation protocol where Alice sends a state ρ_S to Bob, with whom she shares a maximally entangled state $\rho_{AB} = |\Phi_{AB}^+\rangle\langle\Phi_{AB}^+|$. What is Bob's marginal state *before* Alice sends the classical information $x, z \in \{0, 1\}$?

- (a) ρ ,
- (b) $\frac{1}{2}\mathbb{1}$,
- (c) $X^x Z^z \rho Z^z X^x$.

Question 2. Which of the following quantum channels are recoverable?

- (a) $M_A \mapsto V M_A V^\dagger$, where $V: A \rightarrow B$ is an isometry.
- (b) $M_{AB} \mapsto \text{tr}_B[M_{AB}]$.
- (c) $M_A \mapsto (1-p)M_A + \frac{p}{\dim(\mathcal{H}_A)} \text{tr}[M_A]\mathbb{1}_A$, where $p \in (0, 1)$.